

LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Centralized Audit Trail Compliance Engine

1. Architectural Choice

Selected Architecture: Microservices Architecture

The Centralized Audit Trail Compliance Engine uses a Microservices architecture. The platform separates audit-log ingestion and validation, PII detection and masking, cryptographic integrity verification, secure storage, authentication and access control, audit searching, reviewing and exporting, and audit-processing activity into focused services. Each service exposes defined interfaces while hiding its internal implementation.

2. Reasons for Choosing Microservices Architecture

Reason 1 – Independent Scaling

The system can receive large volumes of enterprise event logs and audit searches. Microservices allow high-demand services such as log ingestion and audit search to be scaled independently without scaling the entire platform.

Reason 2 – Fault Isolation and Modularity

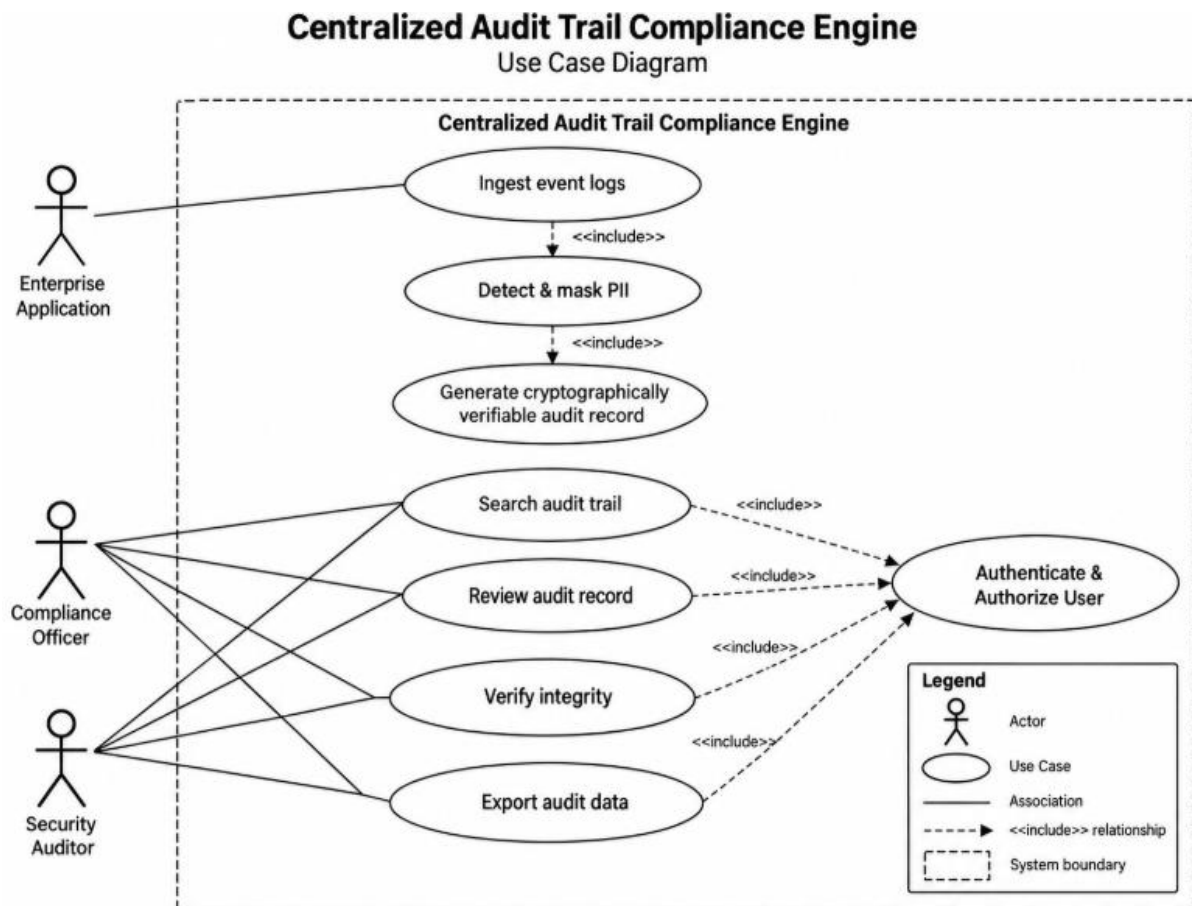
PII masking, integrity verification, storage and reporting are separate responsibilities. A problem in one service can be isolated, diagnosed and redeployed without requiring the complete system to be changed.

3. Security Advantage

Security-sensitive operations can be isolated behind authenticated service interfaces and role-based access control. The PII Detection & Masking Service ensures sensitive information is masked before sanitized audit records are stored, while the Authentication & Access Control Service restricts search, review, verification and export operations to authorized users.

4. Performance Benefit

Log ingestion and audit searching can execute as separate services and scale independently. This supports concurrent processing and high-performance filtering while preventing unrelated functions from becoming a single bottleneck.



Component Diagram Description

The component diagram contains the following major components:

1. **Log Ingestion & Validation Service** – receives enterprise application event logs and validates them before processing.
2. **PII Detection & Masking Service** – identifies and masks sensitive PII such as Social Security Numbers and credit card numbers.
3. **Integrity Verification & Audit Record Service** – generates cryptographic hashes and verifies audit-record integrity.

4. **Secure Audit Log Storage** – stores sanitized audit logs together with integrity information.
5. **Authentication & Access Control Service** – authenticates compliance officers and security auditors and enforces authorization.
6. **Audit Search, Review & Export Service** – supports secure search, review, verification and export of audit information.
7. **Audit Processing Activity Service** – records relevant audit and security processing activity for traceability.

The diagram also includes the Enterprise Applications and Compliance Officer/Security Auditor actors and shows the interfaces, dependencies and interactions between the system components.